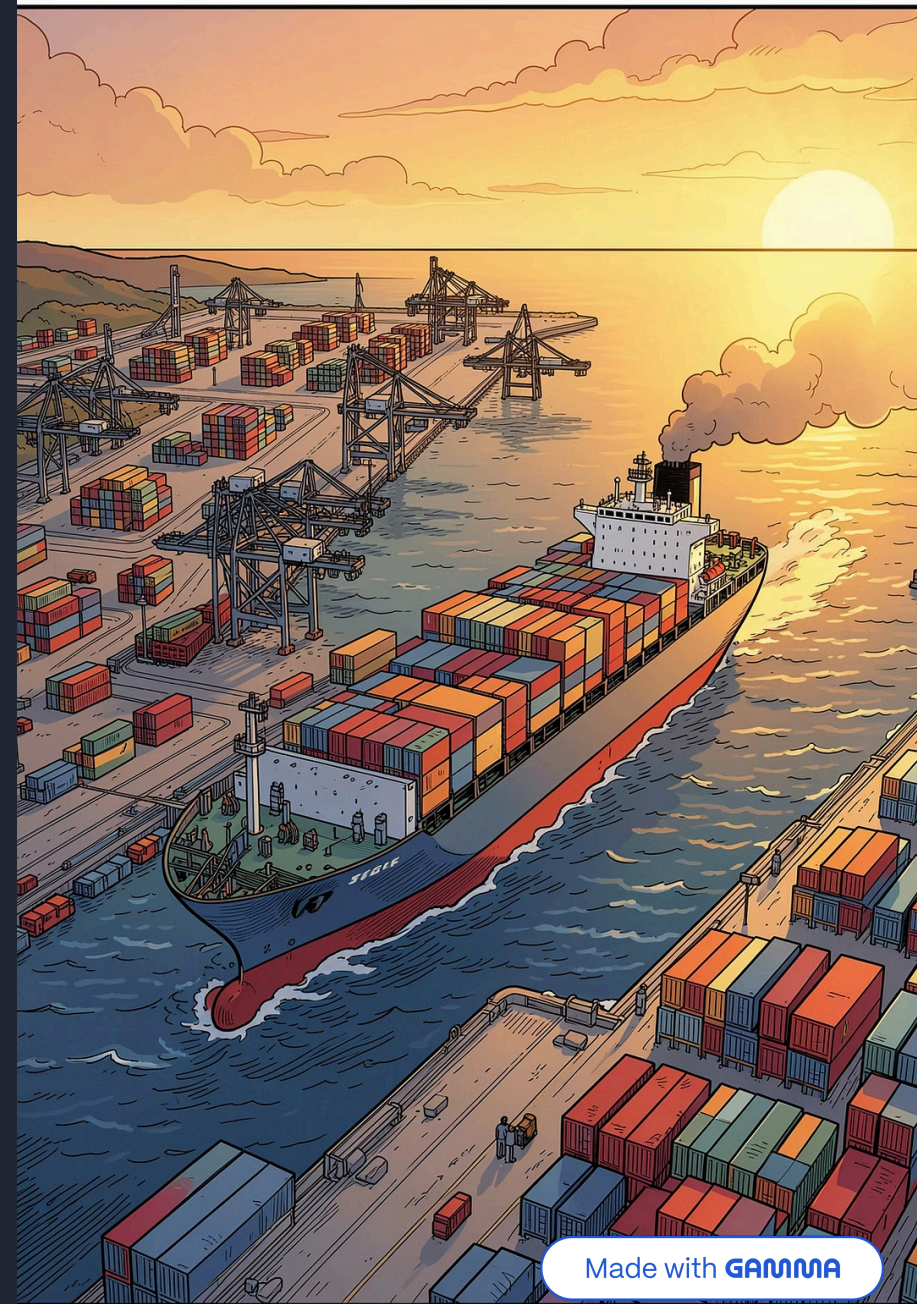


Waze Cargo

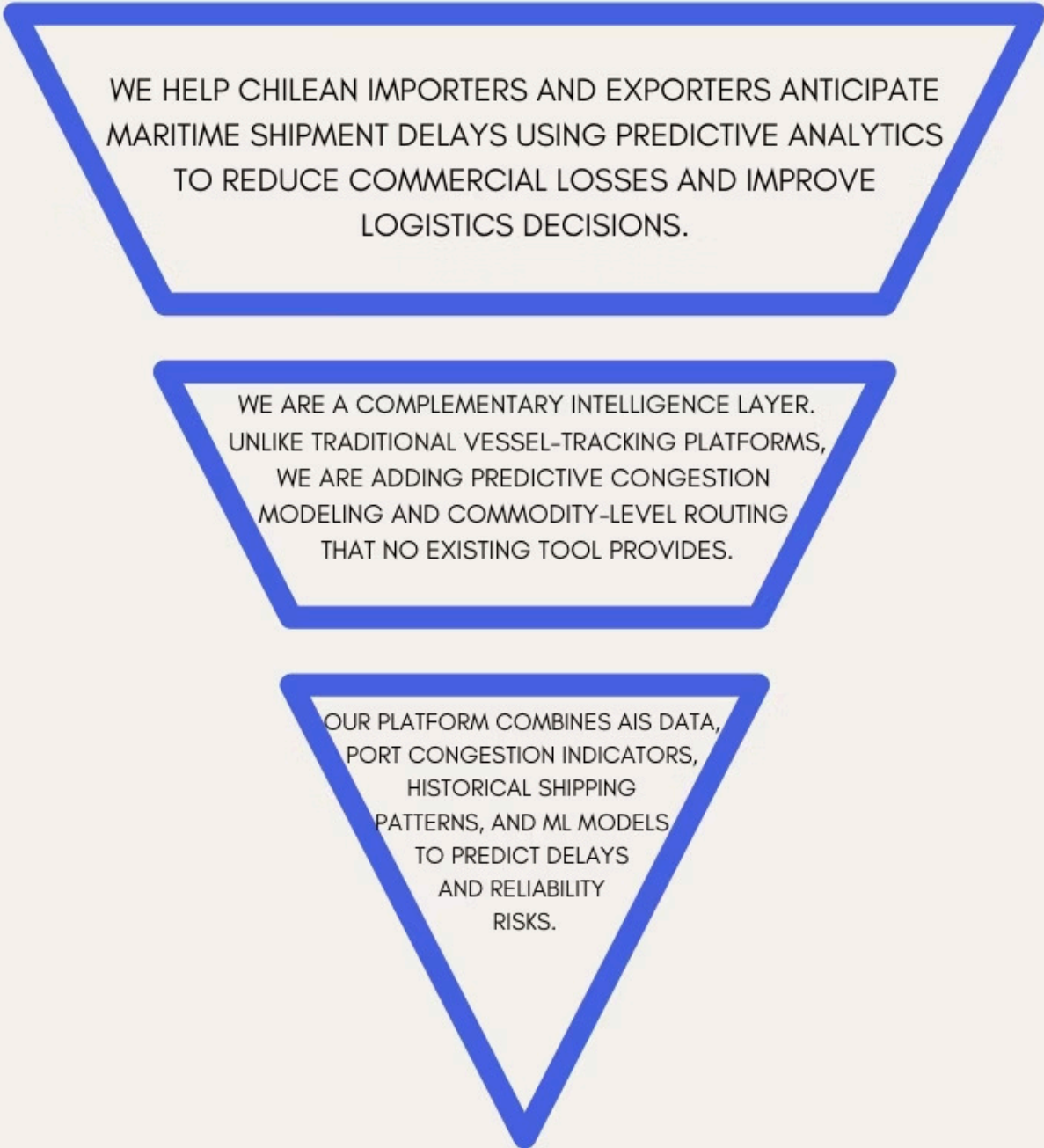
Predictive port congestion intelligence for Chilean importers and exporters — pick the right port, at the right time, before delays happen.

BTS MSC AI & DATA SOLUTIONS — MAY 2026

Meliane Meledje · Michalis Arvanitis · Artemiy Klimkin · Gerson Montesinos · David Vergara



Made with GAMMA



WE HELP CHILEAN IMPORTERS AND EXPORTERS ANTICIPATE
MARITIME SHIPMENT DELAYS USING PREDICTIVE ANALYTICS
TO REDUCE COMMERCIAL LOSSES AND IMPROVE
LOGISTICS DECISIONS.

WHY / VALUE
PROPOSITION

WE ARE A COMPLEMENTARY INTELLIGENCE LAYER.
UNLIKE TRADITIONAL VESSEL-TRACKING PLATFORMS,
WE ARE ADDING PREDICTIVE CONGESTION
MODELING AND COMMODITY-LEVEL ROUTING
THAT NO EXISTING TOOL PROVIDES.

HOW WE'RE
DIFFERENT /
POSITIONING

OUR PLATFORM COMBINES AIS DATA,
PORT CONGESTION INDICATORS,
HISTORICAL SHIPPING
PATTERNS, AND ML MODELS
TO PREDICT DELAYS
AND RELIABILITY
RISKS.

WHAT / HOW
IT WORKS

AARRR Funnel Diagram



01

Importers/exporters discover Waze Cargo via live dashboard, personalised outreach, and trade networks.

Channels: SEO landing page, cold emails/LinkedIn, trade associations.
KPIs: Website visits, Click-Through Rate, signup conversion, CAC.

02

Users see real congestion predictions and reach the “wow” moment quickly.

Channels: In-app walkthroughs, onboarding emails, webinars/conferences.
KPIs: Activation rate, time-to-value, onboarding completion.

03

Users return regularly for forecasts and shipment planning insights.

Congestion alerts, monthly intelligence reports, route history.
KPIs: Retention rate, churn, WAU/MAU, session frequency.

04

Free users upgrade to Pro for detailed insights and routing recommendations.

Channels: In-app upgrade prompts, free trials, upsell emails, seasonal promotions.
KPIs: Free-to-paid conversion, MRR, ARR, ARPU, LTV/CAC.

05

Satisfied users recommend Waze Cargo within logistics and trade networks.

Channels: Referral rewards, WhatsApp sharing, testimonials, trade associations.
KPIs: Referral rate, referral conversion, K-factor, NPS.

GTM Channels & Tactics — AARRR Funnel

Acquisition — How do people first hear about us?

1. Acquisition Strategy

Three discovery routes for Chilean importers and exporters:

- Public live dashboard (no signup required) showing real-time congestion forecasts for San Antonio and Valparaíso
- Personalized cold outreach using 20 years of public Chilean customs data
- Trust-transfer through professional networks and LinkedIn

2. Channel 1: Public Live Dashboard

- Show real-time forecasts on the homepage before registration
- SEO ranking for “*congestión puerto San Antonio*”
- PR through maritime journalists such as Mundo Marítimo and PortalPortuario
- Trust-building for skeptical SME owners
- Removes the “*is this real?*” objection

3. Channel 2: Cold Outreach Using Customs Database

- Hyper-personalized emails referencing actual shipping history
- Example: “*Hi María, your company shipped 14 containers through Valparaíso last March when the port ran 35% above capacity. Would you like the forecast for March 2027?*”
- Cost: **~50 euros/month** for outreach tool
- Position as academic research, not sales, for higher reply rates

4. Future Expansion (Months 4–6)

- Present research at virtual events: ASOEX, SOFOFA, ProChile
- Reach 100–300 decision makers per event
- Requires 50+ active users and case studies first

5. Channels We Skip

- Google Ads — can't outbid Kpler and MarineTraffic
- Physical events in Chile — too expensive for a Barcelona-based team
- Facebook/Instagram ads — not where importers and exporters make professional decisions

6. KPIs Table

KPI	Target	How we measure
Website visits	1,500/month by month 3, 3,000 by month 6	Google Analytics
Click-through rate	Above 3% on cold outreach emails and LinkedIn DMs	Outreach tool and LinkedIn analytics
Visit to signup conversion	Above 10% of visitors create free account	Product analytics
Cost per acquisition	Below 40 euros per free signup	Tool costs plus ad spend divided by new signups

Activation — What is the 'Wowww' Moment?

1. The Wowww Moment

The user checks congestion for a port they actually use and sees a prediction that matches their real-world experience — but with data they've never had before.

- Commodity breakdown showing which products compete for berth space
- Weather risk indicators for swells and wind
- Alternative port recommendation
- **Key insight:** *This is the moment they realize that Waze Cargo gives them visibility no other tool provides.*

2. How We Get Them There Fast

The onboarding sequence is designed to reach the aha moment within minutes, not days.

- **On signup:** onboarding email invites them to track their first shipment
- **In-app walkthrough:** guides them through their first congestion forecast
- **Within one hour of signing up:** personalized welcome email showing congestion forecast for their selected port
- **In-app tooltip or notification:** *"Your port will be 40% above normal in March. Tap to see why."*

3. Channels

- In-app onboarding walkthrough on first login
- Automated email sequence on Day 0, Day 3, and Day 7
- For larger enterprises: live demo webinar

4. KPIs Table

KPI	Target	What it measures
Activation rate	40%	Percentage of users who track their first shipment or check two or more ports in their first week
Time to value	Under 3 minutes	Time from signup to the user seeing their first delay prediction
Onboarding completion rate	70%	Percentage completing the full onboarding walkthrough
First-week engagement	3 or more sessions in week one	Login activity count in the first seven days

Retention — How do we keep them coming back?

Core question: Do importers and exporters come back and keep using Waze Cargo regularly?

1. Why retention is critical for Waze Cargo

Importers and exporters plan shipments weeks and months in advance on tight seasonal cycles:

- Chilean fruit exports: January to April
- Wine: January to March
- Copper: year-round
- **Key insight:** If Waze Cargo delivers accurate forecasts during one season, the natural retention driver is the next season.
- Users return because they trust the model's predictions from last time.
- **Goal:** become the tool they open before every major routing decision, the same way a captain checks weather before departure.

2. Channel 1 — Monthly port congestion alerts

- Automated alerts sent when the ML model detects high congestion risk or adverse weather at a user's frequently used ports
- Example: *"San Antonio forecast: WARNING level swell detected. Consider rerouting to Valparaíso."*
- The ML model already produces a seven-day forecast per port
- Sending alerts when risk level reaches **WARNING** or **CLOSED** is a direct product feature that brings users back without them having to remember to check
- **This is the strongest retention channel for Waze Cargo.**

3. Channel 2 — Monthly trade intelligence digest

- Personalized report summarizing commodity flow trends, port performance, and forecast accuracy for the user's specific HS codes and routes
- Importers and exporters plan monthly
- Fruit exporters check January to April windows, copper exporters plan quarterly shipments
- A monthly summary gives them a reason to re-engage even between active shipments

4. Channel 3 — In-app route history *(future feature)*

- Users will be able to track past routing decisions versus actual congestion outcomes
- Creates a feedback loop that builds trust in the platform
- Also feeds data back into model training
- Requires fully built frontend
- Flagged as future development rather than current retention channel

5. KPIs table

KPI	Target	Rationale
Day 30 and Month 3 retention rate	60% at day 30, 40% at month 3	Day 30 checks if they use it for a real shipment. Month 3 checks if it becomes habitual across a full seasonal cycle.
Churn rate	Below 5% monthly	If a user queries one route and never comes back, the forecast was not useful or trustworthy. Directly measurable.
WAU and MAU	WAU during peak seasons (Jan to Apr), MAU year-round	Nobody checks port congestion daily. The meaningful metric is weekly active users during peak shipping seasons and monthly active users year-round.
Session frequency	At least once per shipment cycle	Realistic and specific to how importers and exporters actually work.
Forecast accuracy satisfaction	Post-shipment feedback survey	If the model predicted HIGH congestion and the user rerouted and saved time, they will come back. If the prediction was wrong, they will not. This is the core retention driver: trust in the model.

Revenue — How do we turn users into paying customers?

The strategy is "taste, then crave."

1. The freemium strategy

The Free tier gives users enough value to build a weekly checking habit, but deliberately locks the features they need to make actual routing decisions. This creates natural upgrade demand without aggressive sales tactics.

2. What Free gives

- Congestion labels for one port showing Low, Moderate, or High status
- Monthly view only, with no weekly or daily granularity
- No commodity detail, just the overall congestion number

3. What Free deliberately locks behind Pro

Three things create the upgrade desire:

- **First:** the commodity table is visible but blurred. Users can see there is a table showing which products compete for berth space, but they cannot read the values.
- **Second:** swell and wind alerts are locked. A banner appears saying *"Swell warning detected — upgrade to see details"* at the exact moment the user needs the information most.
- **Third:** the routing recommendation is blocked. The system shows *"We found a less congested port for your cargo"* but requires Pro to reveal which port and why.

4. Channels

- **In-app upgrade prompts** are the primary channel, working through the blurred commodity table, locked swell alerts, and blocked routing recommendation. These are contextual, not spammy, and are triggered by specific user actions rather than timers.
- **The free trial to paid flow** offers a seven-day Pro trial, but only to users who have checked ports at least twice. This ensures the trial reaches users who already understand the value and are most likely to convert.
- **Upsell emails** reinforce the value gap with personalized messaging. Example: *"You have checked San Antonio eight times this month. Pro users have visibility of all ports and save days per shipment. The sea does not wait. Now you can move before it does."*
- **Promotions** include three mechanics:
 - **Seasonal upgrade push:** one-time Pro preview during peak seasons (grape harvest in March, cherry season in December, copper exports in Q4) with message: *"See what is competing for your port THIS month."*
 - **Annual plan discount:** twelve months for the price of ten, a 17% saving at 1,670 euros per year instead of 2,400.
 - **Trade association group deal:** ASOEX, CNC, and SOFOFA members get 20% off Pro for the first year.

5. KPIs table

KPI	Target	Interpretation
Conversion rate free to paid	8 to 12%	Below 8% means Free is too generous. Above 12% means Free is too restrictive. The sweet spot means the value gap is correctly calibrated.
Monthly Recurring Revenue (MRR)	2,000 euros by month 3, 5,000 by month 6	Measures whether we are growing predictably.
Annual Recurring Revenue (ARR)	40,000 euros by end of year one	The annual run rate that investors and partners evaluate.
Average Revenue Per User (ARPU)	200 euros per month	Below 150 means users pick the cheapest Pro tier. Above 250 means enterprise upsell is working.
Lifetime Value (LTV)	2,400 euros or more	ARPU multiplied by average months retained. At 200 euros per month for 12 months, one user is worth 2,400 euros over their lifetime.
LTV to CAC ratio	Above 3 to 1	If we spend 100 euros to acquire a user, they need to generate 300 euros or more in lifetime value. Below 3 to 1 is unsustainable.

Referral — How do happy customers bring new users?

1. What happens

Importers and exporters operate in tight professional communities. Freight forwarders manage multiple accounts, and trade associations connect hundreds of members. One satisfied user can unlock an entire network.

In maritime logistics, the buyer community is senior and highly interconnected. Port directors, freight forwarders, and cargo operators regularly exchange recommendations at industry events.

Platforms like Windward and Kpler grew significantly through this peer-recommendation dynamic within closed professional networks.

2. Referral channels

Three main channels drive referrals:

- **Invite a logistics partner** is the primary mechanic. Both the sender and the recipient receive one month of Pro for free. The invitation is shared via WhatsApp, which is the primary business communication channel in Chile. A logistics manager forwarding a link with the message *"look at this, it warned me that San Antonio would be congested in March"* carries more weight than any advertisement.
- **Testimonials and case studies** tell real stories of users who avoided delays. Example: *"How my shipments used to have constant delays because of port congestion, and now I can confidently choose a port to receive my cargo from Genoa to San Antonio."*
- **Trade association exposure** involves presenting at ASOEX for exporters, CNC (National Chamber of Commerce), and SOFOFA (industry federation). One presentation reaches multiple potential users within the target segment.

3. KPIs table

KPI	Target	Rationale
Percentage of users who refer	15%	Ambitious but realistic because one trusted voice in maritime logistics carries disproportionate weight. Measured by dividing the number of users who sent at least one successful referral by total active users in the same period.
Referral conversion rate	25%	General SaaS benchmarks place referred-user conversion at 20 to 40%, but maritime cargo intelligence has higher activation friction. We set 25% as a conservative floor. Measured by dividing referred signups who reached first meaningful action by total referred signups.
K-factor	Above 0.3	A K-factor of 0.3 means every 10 users organically bring 3 more. It will not produce viral growth, which requires K above 1, but it meaningfully reduces CAC. Calculated as average invites sent per user multiplied by conversion rate of those invites.
NPS	Above 50	Net Promoter Score measures promoters minus detractors. A score above 50 is considered excellent and is the leading indicator of whether the referral funnel can function at all. Calculated as percentage of promoters minus percentage of detractors.

7. Pricing & Business Model

Revenue model: Freemium SaaS with monthly subscription. The Free tier serves as the acquisition and habit-formation layer. The Pro tier is the core revenue driver. The Enterprise tier is the expansion target for large companies and channel partners.

Price tiers:

Tier	Price	What is included	Target user
Free	0 euros	Congestion labels (Low/Moderate/High) for one port, monthly view only, no commodity detail	User acquisition and habit formation
Pro	100 to 350 euros per month	Full congestion forecast for all ports, commodity breakdown, swell and wind alerts, alternative port recommendations, 12-month forecast, API access	SME importers and exporters with regular trade
Enterprise	Custom, starting at 500 euros per month	Multi-user dashboards, custom ERP and TMS integrations, dedicated support, fleet-level vessel tracking, white-label option for customs brokers	Large importers, freight forwarders, port authorities

Why this model fits the ICP: SMEs in Chile are price-sensitive and skeptical of SaaS products. The Free tier removes all risk from trying the product. The gap between Free and Pro is carefully designed so that Free creates a weekly checking habit while withholding the decision-support features that justify the Pro price. The Pro tier at 100 to 350 euros per month is positioned below MarineTraffic, which charges 100 to 150 euros for tracking only, and well below Kpler at over 500 euros, while delivering more actionable intelligence than either. Most importantly, a single avoided delay of three to five days at approximately 1,500 dollars per day in demurrage pays for two to four months of the Pro subscription. The ROI argument is immediate and concrete.

8. Launch Plan – First 90 Days

Goal linked to NSM: Reach 100 Weekly Active Port Checks with at least 10 paying Pro users generating 2,000 euros in MRR by the end of month three.

Month 1 – Foundation and Free Launch

Deploy the MVP web application using React and FastAPI on AWS EC2 with SSL and a custom domain. Launch the Free tier showing congestion forecasts for San Antonio and Valparaíso, covering both import and export directions. Publish four LinkedIn posts featuring port congestion insights to build authority and generate initial traffic. Reach out to three importers and exporters through the team's Chilean network and onboard them as beta partners with free access and a broker dashboard. Target: 50 free signups with 20 activated users who check two or more ports in their first week.

Month 2 – Activation and Pro Beta

Launch the weekly congestion digest email for all free users to drive retention. Enable swell and wind alerts for 10 selected active users as Pro beta testers. Run the first cold outreach campaign using customs data for hyper-personalised emails. Host the first live demo webinar in Spanish, targeting 30 attendees. Target: 100 total signups, 40 activated users, and 10 Pro beta users testing the full feature set.

Month 3 – Revenue and Referral

Launch the Pro tier publicly at 100 to 350 euros per month with a seven-day free trial for activated users. Activate the referral programme where both sender and receiver get one month of Pro free via WhatsApp invitation. Present at one trade association event, either ASOEX or a CNC chapter meeting. Publish the first case study featuring a beta Pro user who avoided a delay. Target: 150 total signups, 100 Weekly Active Port Checks, 10 paying Pro users, and 2,000 euros in MRR.

9. Risks & Assumptions

Biggest assumptions in this GTM:

We assume that Chilean SME importers and exporters will adopt a SaaS tool for logistics decisions. We assume the Free tier creates enough value to form a weekly habit but not enough to eliminate the need for Pro. We assume that the model's R-squared accuracy of 0.909 on historical data will translate to actionable accuracy on future predictions. We assume Open-Meteo weather data is reliable enough for port closure predictions along the Chilean coast.

Main risks and mitigation:

Risk	Severity	How we mitigate
Low SaaS adoption among Chilean SMEs	High	Free tier removes cost barrier. WhatsApp integration for alerts meets users where they already communicate.
Competitors add prediction features	Medium	Our moat is 20 years of Chilean customs data with commodity granularity. Global platforms will not build Chile-specific models for a niche within a 1.8 billion dollar global market.
ML accuracy degrades on future data	Medium	Walk-forward cross-validation designed to catch this. Monthly auto-retrain. Monitor prediction versus actual and alert if R-squared drops below 0.7.
Weather API reliability (Open-Meteo)	Low	Free API with no SLA. Fallback: SHOA buoy data as backup source. Consider paid marine weather API such as Stormglass for the Enterprise tier.
Customs data stops being publicly available	Low	SNA has published data since 2002 with no interruption. Risk is minimal but mitigation includes pursuing a data partnership with SNA or licensed data providers.
Team capacity – small team, large scope	High	Focus on two ports only for the MVP: San Antonio and Valparaíso. Expand to more ports only after product-market fit is confirmed with the first 50 paying users.